

RAMNICU SARAT, Romania

WMO: 153070

Lat: 45.3906N

Lon: 27.0386E

Elev: 505

StdP: 14.43

Time Zone: 2.00 (EUE)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	13.4	16.6	7.4	8.2	16.9	10.8	9.7	20.0	26.6	29.6	21.1	27.2	4.7	0	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.8	90.8	70.8	88.1	69.9	85.8	69.1	74.5	85.4	72.6	83.7	71.1	82.2	5.8	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.5	118.4	80.0	69.1	109.1	77.5	67.4	102.5	76.0	38.3	85.6	36.5	83.4	35.2	82.5	84.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.5	15.0	12.5	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	53.2	30.3	36.4	43.2	52.3	63.0	71.0	72.9	73.5	64.2	52.8	43.1	34.6		
	DBStd	16.64	7.67	8.47	8.60	7.76	5.95	5.23	4.98	5.78	7.20	7.74	8.26	7.96		
	HDD50	2084	612	387	240	67	2	0	0	0	0	60	239	477		
	HDD65	5151	1077	802	675	385	106	12	4	7	102	381	658	942		
	CDD50	3246	0	5	30	135	405	629	709	728	426	147	31	1		
	CDD65	838	0	0	0	3	45	191	248	270	78	3	0	0		
	CDH74	7286	0	0	3	46	383	1563	2161	2460	640	30	0	0		
	CDH80	2501	0	0	0	5	72	473	778	983	188	2	0	0		
	WSAvg	5.1	4.7	5.3	6.0	5.9	5.3	5.5	5.1	5.3	4.8	4.8	4.3	4.9		
	PrecAvg	21.2	1.1	0.9	1.2	1.7	2.6	2.8	2.4	1.9	2.0	1.9	1.3	1.4		
(15) Precipitation	PrecMax	30.9	2.7	2.4	2.7	4.1	6.8	5.2	6.0	4.6	4.1	6.4	3.8	3.7		
	PrecMin	14.4	0.0	0.1	0.0	0.8	0.5	0.4	0.2	0.3	0.4	0.1	0.0	0.0		
	PrecStd	4.6	0.7	0.7	0.7	0.8	1.6	1.1	1.4	1.2	1.1	1.4	1.0	1.1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.4	63.9	72.6	79.0	87.8	90.7	94.0	94.8	89.7	78.6	68.6	56.7		
		MCWB	46.7	53.3	56.6	61.5	69.7	72.5	75.6	71.6	67.2	64.3	56.9	49.1		
	2%	DB	48.2	58.0	67.1	73.7	82.0	87.6	90.3	91.1	84.6	73.2	63.3	52.6		
		MCWB	42.7	50.6	53.5	59.0	65.4	71.2	72.5	69.4	65.0	61.5	54.1	46.4		
	5%	DB	43.8	53.4	61.5	69.7	78.4	85.2	87.3	88.4	80.8	69.0	58.5	48.7		
		MCWB	39.8	46.7	50.5	56.6	63.9	70.1	70.6	68.9	63.8	58.7	52.0	44.5		
	10%	DB	40.5	49.1	56.8	66.1	75.4	82.5	84.5	85.7	76.6	65.0	54.5	45.2		
		MCWB	37.9	44.2	48.0	54.9	62.4	69.0	69.2	68.4	62.8	56.7	49.9	41.2		
	0.4%	WB	47.1	55.3	59.8	62.7	70.7	76.0	78.5	75.6	70.3	67.1	58.2	51.1		
		MCDB	52.7	62.4	70.4	77.5	81.4	85.0	87.7	88.3	82.5	75.0	67.5	54.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	43.2	50.8	54.2	60.1	67.5	73.5	75.4	73.1	67.5	63.2	55.3	48.0		
		MCDB	46.8	57.3	64.4	72.2	78.0	84.8	86.2	85.8	77.6	69.9	61.5	52.1		
	5%	WB	40.5	47.4	51.2	57.9	65.7	71.8	72.9	71.4	65.9	60.3	53.1	45.1		
		MCDB	43.2	52.6	60.2	68.1	75.5	83.0	82.4	84.6	77.9	66.7	57.8	49.1		
	10%	WB	38.0	44.6	48.7	55.7	64.1	70.1	71.1	69.9	64.0	58.0	50.6	42.0		
		MCDB	40.1	48.4	55.6	64.5	72.9	81.1	81.0	81.9	75.4	63.8	54.2	45.3		
	5% DB	MDBR	12.0	13.9	16.1	17.7	18.4	18.3	18.8	19.5	18.2	15.8	12.2	11.4		
		MCDBR	17.7	21.2	24.9	25.0	22.8	21.8	23.6	23.0	23.4	21.8	18.2	16.2		
		MCWBR	12.8	14.2	14.3	13.3	10.5	9.6	9.0	8.1	9.3	11.9	11.1	10.9		
		MCWBR	15.7	19.6	23.2	23.8	20.3	20.5	20.1	20.4	20.9	20.3	16.6	15.2		
(35) Mean Daily Temperature Range	5% WB	MCWBR	11.7	14.1	14.0	13.0	10.5	10.1	8.8	8.4	9.2	12.2	10.4	10.7		
	taub		0.328	0.357	0.392	0.451	0.425	0.427	0.442	0.456	0.416	0.380	0.353	0.328		
		taud	2.337	2.255	2.215	2.072	2.226	2.256	2.219	2.170	2.246	2.331	2.358	2.347		
(42) Clear-Sky Solar Irradiance	Ebn at Noon		243	256	263	256	268	268	262	252	252	245	231	228		
		Edh at Noon	25	33	39	49	43	42	43	44	37	30	24	23		
	RadAvg		412	673	1039	1424	1791	1903	1938	1750	1271	826	453	347		
(45) All-Sky Solar Radiation	RadStd		51	81	108	147	143	142	141	118	111	93	44	44		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (59 neighbors)	+0.95	N/A	N/A	N/A	N/A	N/A	N/A	-205	-272	+122	+43

Nomenclature: See separate page